THIRUVENKADAPRASATH.S

192312016

9	Compare Linear and Polynomial Regression using Python

CSV FILE

```
Position_Level,Salary
1,45000
2,50000
3,60000
4,80000
5,110000
6,150000
7,200000
8,300000
9,500000
10,1000000
```

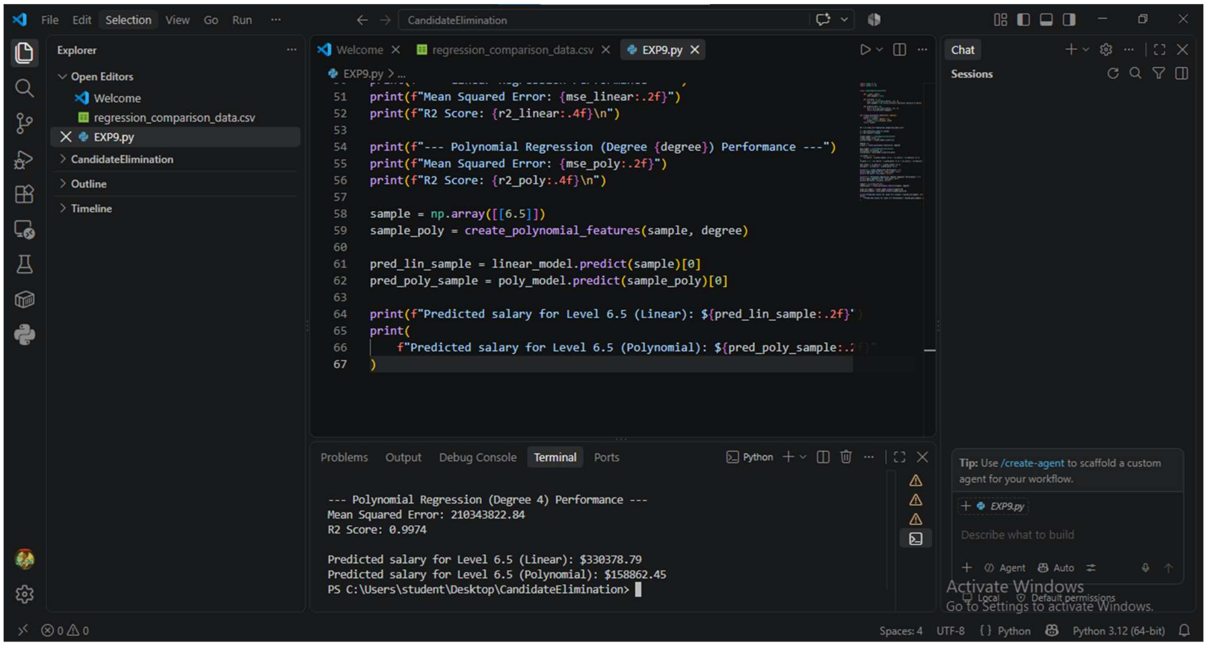
PROGRAM

```
import numpy as np
import pandas as pd



class LinearRegressionScratch:

    def __init__(self):
        self.weights = None

    def fit(self, X, y):
        X_b = np.c_[np.ones((len(X), 1)), X]
```

```python
        self.weights = np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)

    def predict(self, X):
        X_b = np.c_[np.ones((len(X), 1)), X]
        return X_b.dot(self.weights)



def create_polynomial_features(X, degree):
    X_poly = X.copy()
    for d in range(2, degree + 1):
        X_poly = np.c_[X_poly, X**d]
    return X_poly



df = pd.read_csv("regression_comparison_data.csv")

X = df[["Position_Level"]].values
y = df["Salary"].values

linear_model = LinearRegressionScratch()
linear_model.fit(X, y)
y_pred_linear = linear_model.predict(X)

degree = 4
X_poly = create_polynomial_features(X, degree)

poly_model = LinearRegressionScratch()
poly_model.fit(X_poly, y)
y_pred_poly = poly_model.predict(X_poly)
```

```python
r2_linear = 1 - (
    np.sum((y - y_pred_linear) ** 2) / np.sum((y - np.mean(y)) ** 2)
)
r2_poly = 1 - (np.sum((y - y_pred_poly) ** 2) / np.sum((y - np.mean(y)) ** 2))

mse_linear = np.mean((y - y_pred_linear) ** 2)
mse_poly = np.mean((y - y_pred_poly) ** 2)

print(f"--- Linear Regression Performance ---")
print(f"Mean Squared Error: {mse_linear:.2f}")
print(f"R2 Score: {r2_linear:.4f}\n")

print(f"--- Polynomial Regression (Degree {degree}) Performance ---")
print(f"Mean Squared Error: {mse_poly:.2f}")
print(f"R2 Score: {r2_poly:.4f}\n")

sample = np.array([[6.5]])
sample_poly = create_polynomial_features(sample, degree)

pred_lin_sample = linear_model.predict(sample)[0]
pred_poly_sample = poly_model.predict(sample_poly)[0]

print(f"Predicted salary for Level 6.5 (Linear): ${pred_lin_sample:.2f}")
print(
    f"Predicted salary for Level 6.5 (Polynomial): ${pred_poly_sample:.2f}"
)
```

OUTPUT



print(f"Mean Squared Error: {mse_linear:.2f}")
print(f"R2 Score: {r2_linear:.4f}\n")
print(f"--- Polynomial Regression (Degree {degree}) Performance ---")
print(f"Mean Squared Error: {mse_poly:.2f}")
print(f"R2 Score: {r2_poly:.4f}\n")
sample = np.array([[6.5]])
sample_poly = create_polynomial_features(sample, degree)
pred_lin_sample = linear_model.predict(sample)[0]
pred_poly_sample = poly_model.predict(sample_poly)[0]
print(f"Predicted salary for Level 6.5 (Linear): ${pred_lin_sample:.2f}"
print(
f"Predicted salary for Level 6.5 (Polynomial): ${pred_poly_sample:.
)
--- Polynomial Regression (Degree 4) Performance ---
Mean Squared Error: 210343822.84
R2 Score: 0.9974
Predicted salary for Level 6.5 (Linear): $330378.79
Predicted salary for Level 6.5 (Polynomial): $158862.45
PS C:\Users\student\Desktop\CandidateElimination>